Semantic Coordinate Identity Tokenization (SCIT)

Semantic Coordinates, Identity-Native Representation, and the Reorganization of Machine Cognition Around Governed Meaning Address Spaces

Where Language Names Meaning, Coordinates Locate It

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Abstract

Large language models currently operate over tokenization systems optimized around lexical recurrence, statistical compression, and surface-level representational efficiency. While such tokenization schemes achieve remarkable practical performance, they remain fundamentally organized around linguistic artifacts rather than semantic identities.

As a result, reasoning systems repeatedly reconstruct semantic distinctions that are conceptually stable yet lexically ambiguous. A single token may correspond to multiple word senses, conceptual regimes, or epistemic jurisdictions. Semantic disambiguation therefore becomes an ongoing computational burden repeatedly paid throughout inference.

This paper introduces Semantic Coordinate Identity Tokenization (SCIT), a representation framework in which machine cognition operates over semantic coordinates rather than lexical surfaces. SCIT extends the Semantic Crystallization (SemCrys) program by proposing that stabilized semantic structures should not merely become tokenizer-native symbols, but should be assigned governed semantic coordinates that simultaneously function as identity, address, and topological location within a semantic substrate.

SCIT distinguishes between semantic identity and semantic surface. A semantic coordinate is not merely a label attached to meaning. Rather, it is a position within a governed semantic topology whose relationships, exclusions, adjacencies, and disambiguation pathways are explicitly represented.

The framework introduces the Semantic Coordinate Identity Layer (CSIL), Global Term Key (GTK), semantic coordinate spaces, coordinate-indexed representation, semantic addressability, and topology-aware tokenization. It argues that future

reasoning systems may benefit from operating over governed semantic coordinates rather than repeatedly reconstructing meaning from lexical fragments.

SCIT does not propose the replacement of natural language. Rather, it proposes a semantic substrate beneath language through which concepts become directly addressable cognitive objects.

Importantly, SCIT does not primarily seek token-count reduction. Its objective is reduction of amortized semantic reconstruction cost through the stabilization of recurring semantic identities within governed semantic infrastructure.

The central claim of SCIT is simple:

> Meaning may become computationally more efficient when concepts are represented by governed semantic coordinates rather than reconstructed repeatedly from lexical ambiguity.

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1. Introduction

Modern language models operate through a remarkable sequence of transformations.

- Natural language surfaces are converted into token sequences.
- Token sequences are converted into embeddings.
- Embeddings are transformed through attention operations.
- Reasoning emerges through interactions among these representational objects.

Yet despite the sophistication of contemporary architectures, an important representational asymmetry remains:

- Language is organized around words.
- Thought is not.

Many recurring conceptual structures possess stable semantic identities while remaining lexically ambiguous. Consider:

- authority
- governance
- legitimacy
- control
- responsibility

Each term admits multiple distinct meanings depending upon context. The token may remain stable. The semantic identity does not. Reasoning systems therefore repeatedly spend computational effort reconstructing which meaning is active.

SCIT proposes a different approach.

Rather than treating lexical surfaces as primary and semantic identities as inferred, SCIT treats semantic coordinates as primary and lexical surfaces as projections. More specifically, SCIT shifts a substantial portion of ambiguity resolution from runtime inference toward semantic infrastructure, coordinate assignment, topology maintenance, and governance.

SCIT externalizes semantic disambiguation into a persistent, governed substrate, converting repeated inference-time ambiguity resolution into amortized infrastructure cost. This redistribution of semantic work constitutes one of the framework's central claims.

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2. The Coordinate Principle

The foundational insight of SCIT is that semantic identity and semantic location are inseparable.

Traditional identifiers function primarily as labels. For example, `c_auth:legit_binding` may be assigned a unique coordinate. However, the identifier alone contains no information about its semantic neighborhood. Its relationship to adjacent concepts remains external.

SCIT proposes a different model. A semantic coordinate exists within a governed semantic topology. The coordinate simultaneously functions as:

- identity
- address
- location
- reference point

The coordinate is therefore not merely attached to meaning. The coordinate is a topologically grounded identity function over semantic space.

2.1 Formal Definition of Semantic Coordinates

Let semantic space be defined as

$$S = (V, E, \tau, \gamma)$$

where:

- V is the set of semantic coordinates.
- $E \subseteq V \times V$ is the set of semantic relations.
- $\tau : E \to T$ assigns relation types.
- $\gamma : V \to \mathcal{M}$ maps coordinates to governance metadata.

The graph E may be directed. Some relation types may additionally declare symmetry, but symmetry is relation-typed rather than globally assumed.

Optional edge weighting may be defined as

$$w : E \to \mathbb{R}$$

and relation-type weighting may be defined as

$$w_\tau : T \to \mathbb{R}$$

with

$$w(c, c') = w(e) \quad \text{where} \quad e = (c, c')$$

A semantic coordinate $c \in V$ is not merely a unique identifier. It is a node embedded within a governed semantic topology. Identity therefore emerges from structure.

2.2 Neighborhood Definition

Neighborhoods must be explicitly defined. A typed neighborhood is

$$\mathcal{N}(c) = \{(c', \tau(e)) \mid e = (c, c') \in E\}$$

A k-hop neighborhood is

$$\mathcal{N}_k(c)$$

representing all reachable coordinates within k typed traversals. Neighborhoods may additionally incorporate weighted edge contributions through $w(e)$.

For tractability, coordinate degree may be bounded

$$|\mathcal{N}(c)| \leq d_{max}$$

This prevents topology explosion and stabilizes complexity.

All SCIT inference operates under bounded locality in semantic topology:

$$k \leq k_{max}, \quad h \leq h_{max}$$

This locality assumption prevents implicit global graph traversal and makes coordinate resolution computationally tractable.

2.3 Topological Distance

SCIT relies on a topological distance function. One minimal formulation is

$$d_{topo}(c_i, c_j) = \alpha \cdot d_{path}(c_i, c_j) + \beta \cdot d_{neigh}(\mathcal{N}_k(c_i), \mathcal{N}_k(c_j))$$

where d_{path} is shortest-path distance over typed edges, d_{neigh} is distributional divergence over neighborhood types, and α, β are weighting coefficients.

In the unweighted case, d_{path} may be implemented as shortest hop count. In the typed case,

$$d_{path}(c_i, c_j) = \sum_{e \in path(c_i, c_j)} w_{\tau(e)}$$

where $w_{\tau(e)}$ is a relation-type cost.

Neighborhood type distribution may be defined as

$$P_{\mathcal{N}_k(c)}(t) = \frac{|\{e \in \mathcal{N}_k(c) : \tau(e) = t\}|}{|\mathcal{N}_k(c)|}$$

with normalization

$$\sum_t P_{\mathcal{N}_k(c)}(t) = 1$$

If

$$|\mathcal{N}_k(c)| = 0$$

the system may use a uniform prior, skip d_{neigh}, or fall back to embedding-only distance.

Neighborhood divergence may be instantiated as

$$d_{neigh} = JS\big(P_{\mathcal{N}_k(c_i)}, P_{\mathcal{N}_k(c_j)}\big)$$

where JS denotes Jensen-Shannon divergence.

This makes identity stability, admissibility, and topology-aware retrieval computable rather than merely descriptive.

2.4 Identity Stability

Exact neighborhood equivalence is unrealistic in evolving semantic systems. SCIT therefore defines identity through stability under bounded perturbation.

Define

$$c_i \sim c_j \iff d_{topo}(c_i, c_j) < \epsilon$$

Identity persistence therefore depends upon topological continuity rather than exact graph invariance. When many coordinates satisfy this relation, they may be collected into an equivalence class

$$[c] = \{c' \mid c' \sim c\}$$

Governance may then choose to merge, preserve, fork, or namespace-distinguish the cluster depending on jurisdictional requirements.

For any coordinate under admissible updates, long-term identity persistence should satisfy

$$\mathbb{E}\big[d_{topo}(c^{(t)}, c^{(t+1)})\big] < \delta$$

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3. Three Representational Spaces

SCIT distinguishes three distinct representational domains.

- Lexical space L: natural-language expressions and symbolic surfaces.
- Semantic coordinate space V: governed semantic identities.
- Embedding space $\mathbb{R}^n$: continuous learned representations.

Mappings are

$$L \to V \to \mathbb{R}^n$$

with optional inverse projection

$$\mathbb{R}^n \to V$$

for coordinate recovery and semantic decoding. Maintaining separation among these spaces prevents conceptual collapse between symbolic, lexical, and continuous representations.

3.1 Cross-Lingual Projection

SCIT treats language as projection over coordinate identity. Let

$$\pi : L \to V$$

map lexical expressions into semantic coordinates. For semantically equivalent expressions across languages,

$$\forall l_1, l_2 \in L, \quad \pi(l_1) = \pi(l_2) = c$$

when l_1 and l_2 express the same governed semantic identity. Cross-lingual invariance therefore attaches to the coordinate, not to surface form.

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4. From Tokens to Coordinates

4.1 Lexical Tokenization

Contemporary tokenizers operate over:

- character sequences
- lexical recurrence
- byte-pair compression
- unigram statistics

These systems optimize representational efficiency at the surface layer. The resulting token may already be highly compressed. For example, `authority` may exist as a single token. From the tokenizer's perspective, no additional compression opportunity exists. Yet the semantic burden remains.

The model must still determine:

- authority as expertise
- authority as institutional office
- authority as legitimate binding power
- authority as source citation
- authority as coercive domination

The lexical token remains singular. The instantiated semantic identity does not.

4.2 Coordinate-Indexed Tokenization Interface

SCIT proposes that tokenization may instead occur over semantic coordinates through a coordinate-indexed tokenization interface.

Conceptually:

- Natural Language
- ↓ GTK Resolution
- ↓ CSIL Coordinate
- ↓ SCIT
- ↓ Embedding
- ↓ Model

The tokenizer no longer receives lexical surfaces as its primary object. Instead, it receives semantic coordinates whose identities have already been resolved. The objective is not character compression. The objective is semantic stabilization.

4.3 Example Resolution Walkthrough

Input:

> The authority of the court was challenged.

GTK generates candidate coordinates:

- `c_auth:expertise`
- `c_auth:institutional_office`
- `c_auth:institutional_legitimacy`

Context activates:

- court
- challenge
- legitimacy
- jurisdiction

GTK evaluates neighborhood compatibility. The selected coordinate becomes:

- `c_auth:institutional_legitimacy`
  Attention subsequently operates over semantically resolved concepts rather than ambiguous lexical surfaces.

4.4 End-to-End Execution Trace

The minimal execution trace is

$$
\begin{aligned}
& x \\
& \rightarrow C_x \\
& \rightarrow C_{context} \\
& \rightarrow c^* \\
& \rightarrow \phi(c^*)
\end{aligned}
$$

Instantiated:

$$C_x = \text{Gen}(x, L, V)$$

$$C_{context} = \text{Context}(x)$$

$$c^* = \arg\max_{c \in C_x} P(c \mid x, \mathcal{N}(c), G, C_{context})$$

$$\phi(c^*) \in \mathbb{R}^n$$

This trace is the operational core of SCIT: lexical input is candidate-generated, contextualized, topology-scored, governance-filtered, resolved to coordinate identity, and realized as an embedding. The execution trace defines a closed computational pathway from lexical input to coordinate-resolved embedding, ensuring that all intermediate transformations are explicitly governed, bounded, and composable.

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5. The Semantic Coordinate Identity Layer

5.1 CSIL

The Semantic Coordinate Identity Layer (CSIL) functions as the coordinate registry layer of the architecture. Each coordinate represents a disambiguated semantic identity. The coordinate remains invariant across:

- languages
- symbol systems
- tokenizers
- projections
- representational surfaces

The coordinate therefore functions as a semantic anchor independent of expression.

5.2 Registry and Runtime Graph

CSIL is the canonical coordinate registry. Runtime reasoning does not require loading the entire registry. Instead, systems may operate over an active semantic view

$$S_{active} \subseteq S$$

where S_{active} is scoped by:

- domain
- jurisdiction
- task
- model capacity
- governance policy
- retrieval context

More explicitly,

$$S_{active} = \{c \in V \mid G(c) > 0 \wedge Scope(c)\}$$

This separates canonical identity storage from runtime semantic computation.

5.3 Namespaces and Collision Constraints

Coordinates must operate across domains, jurisdictions, and semantic traditions. Define namespace function

$$\nu : V \to Namespace$$

Coordinate uniqueness is enforced within namespace:

$$\nu(c_i) = \nu(c_j) \wedge id(c_i) = id(c_j) \Rightarrow c_i = c_j$$

This allows multiple governed semantic domains to coexist without requiring premature global ontology closure.

5.4 Coordinate Admissibility

Not every semantic structure should receive a coordinate. A candidate instantiated semantic identity s may be assigned coordinate c iff:

- Persistence: s remains invariant across contexts above threshold θ_p.
- Distinguishability: there exists coordinate c' such that $d(c, c') > \theta_d$.
- Topological integration: $|\mathcal{N}(c)| \geq k_{min}$
- Governance admissibility: the candidate passes G_{policy}.

These constraints prevent semantic fragmentation and establish a promotion pathway from SemCrys crystallization into SCIT coordinates.

5.5 Coordinate Growth Bound

Coordinate proliferation is a structural risk. SCIT therefore requires either an explicit coordinate budget

$$|V| \leq B$$

or a maintenance penalty

$$C_{maintenance} \propto |V|$$

Coordinate promotion should be tied to reuse frequency, stability, distinguishability, and governance admissibility. A coordinate should exist only when its expected amortized reuse exceeds its expected maintenance cost.

5.6 Revision and Fork Semantics

Semantic systems evolve. Coordinates therefore require governed update mechanisms.

Topology itself is time-indexed

$$S^{(t)} = (V^{(t)}, E^{(t)}, \tau^{(t)}, \gamma^{(t)})$$

Define coordinate update

$$c^{(t+1)} = \text{update}(c^{(t)}, \Delta G, \Delta E)$$

Identity persists iff

$$d_{topo}(c^{(t)}, c^{(t+1)}) < \delta$$

If this threshold is exceeded, mutation becomes inadmissible and a coordinate fork is required

$$c' \leftarrow fork(c)$$

This preserves semantic lineage while preventing identity drift.

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6. Semantic Topology

Semantic topology becomes a first-class object within SCIT. These metrics parameterize the scoring functions defined in Section 7.

6.1 Local Density

$$\rho(c) = \frac{|\mathcal{N}(c)|}{|V|}$$

6.2 Path Distance

$$d(c_i, c_j)$$

computed through typed shortest paths.

6.3 Neighborhood Entropy

$$H(c) = -\sum_{t \in T} P(t \mid \mathcal{N}(c)) \log P(t \mid \mathcal{N}(c))$$

6.4 Semantic Curvature

Semantic curvature is defined as deviation of local connectivity from expected connectivity under a selected null model. Future implementations may additionally employ graph-Ricci formulations.

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7. Global Term Key

Coordinates alone are insufficient. A coordinate without topology becomes merely another identifier system. SCIT therefore depends upon Global Term Key (GTK).

GTK supplies:

- disambiguation pathways
- semantic neighborhoods
- synonym structures
- exclusion sets
- adjacency relationships
- semantic governance constraints

Meaning is inferred via constrained traversal under governance-weighted topology.

7.1 Governance Functions

Governance must be operational rather than philosophical. Define

$$G : V \cup E \to [0, 1]$$

Governance may function as:

- admissibility filter
- ranking prior
- mutation gate
- promotion criterion
- topology constraint

Governance therefore participates directly in semantic resolution. Governance may also prune topology itself

$$E' = \{e \in E \mid G(e) > 0\}$$

so that inadmissible relations are removed from the active reasoning graph rather than merely downweighted.

A hard admissibility rule may be defined as

$$P(c \mid x, \dots) = 0 \quad \text{if} \quad G(c) = 0$$

A soft governance contribution may be defined as

$$s_G(c) = \log G(c) + \sum_{e \in \mathcal{N}(c)} \log G(e)$$

when $G(c) > 0$ and edge governance values are nonzero. This prevents governance from remaining merely advisory and ensures edge-level governance participates in inference.

7.2 Candidate Generation

SCIT separates candidate generation from coordinate selection. Given input x, candidate coordinates are generated by

$$C_x = \text{Gen}(x, L, V)$$

with bound

$$|C_x| \leq k_{max}$$

Candidate generation may use:

- lexical priors
- string-to-coordinate maps
- embedding nearest neighbors
- coordinate description retrieval
- multilingual projection tables

The resolver then selects or weights candidates using topology and governance.

If

$$C_x = \varnothing$$

the system may either revert to the lexical token pathway or assign a provisional frontier distribution

$$c \sim P(V_{proto})$$

7.3 Context Construction

Resolution depends on context coordinates. Define

$$C_{context} = \text{Context}(x)$$

where $C_{context}$ may be constructed from:

- neighboring resolved coordinates
- sliding-window lexical projections
- prior disambiguations
- retrieved coordinate neighborhoods
- multimodal context signals

Context must be bounded

$$|C_{context}| \leq k_{ctx}$$

or decay-weighted by distance

$$w(c, c') \propto e^{-\lambda \cdot \Delta position}$$

This prevents unbounded context accumulation.

7.4 Iterative Context Refinement

Resolution may be iterative rather than single-pass. Define

$$C_{context}^{(t+1)} = \text{Update}(C_{context}^{(t)}, C_x)$$

Iteration terminates when

$$C_{context}^{(t+1)} = C_{context}^{(t)}$$

or when a fixed maximum iteration count T is reached. This allows earlier candidate resolutions to inform later disambiguation, and later constraints to revise earlier uncertain assignments without permitting unbounded oscillation. Such refinement is especially important for compositional sentences where meaning emerges across phrase boundaries.

7.5 Disambiguation as Topology-Constrained Inference

Given input x, candidate coordinates C_x, and context $C_{context}$,

$$C_x^{valid} = \{c \in C_x \mid G(c) > 0\}$$

$$c^* = \arg \max_{c \in C_x^{valid}} P(c \mid x, \mathcal{N}(c), G, C_{context})$$

where:

- x is lexical, symbolic, or multimodal input.
- $\mathcal{N}(c)$ is local semantic topology.
- G represents governance constraints.
- $C_{context}$ represents resolved or projected contextual coordinates.

Disambiguation therefore becomes an optimization process operating over semantic neighborhoods rather than a dictionary lookup procedure.

A modular scoring formulation is

$$s(c) = s(x, c) + s_{topo}(c, C_{context}) + s_G(c)$$

$$P(c \mid x, \mathcal{N}(c), G, C_{context}) = \frac{\exp(s(c))}{\sum_{c' \in C_x^{valid}} \exp(s(c'))}$$

where:

- $s(x, c)$ captures lexical or multimodal fit.
- $s_{topo}(c, C_{context})$ captures neighborhood coherence.
- $s_G(c)$ captures governance weighting.

A context-aggregated topological distance may be defined as

$$d_{topo}(c, C_{context}) = \frac{1}{|C_{context}|} \sum_{c' \in C_{context}} d_{topo}(c, c')$$

A unified topology score may then be defined as

$$s_{topo}(c, C_{context}) = -\lambda_d \cdot d_{topo}(c, C_{context}) + \lambda_a \cdot \sum_{c' \in C_{context}} w(c, c')$$

where $w(c, c')$ reflects typed adjacency strength.

7.6 Uncertainty Representation

SCIT does not require hard assignment. Coordinate activation may remain probabilistic:

$$P(c \mid x)$$

Resolution uncertainty becomes:

$$H(C_x)$$

Multiple coordinates may remain simultaneously active when ambiguity remains unresolved. SCIT therefore does not eliminate probabilistic reasoning. It externalizes part of the semantic disambiguation burden into a persistent governed substrate while preserving uncertainty where required.

Under stable coordinate regimes, the ambiguity-reduction invariant is

$$\sum_i H(C_{x_i})_{SCIT} \leq \sum_i H(C_{x_i})_{lexical}$$

for resolved sequence $x_1, \ldots, x_n$. This is an evaluable claim rather than a guaranteed universal result.

7.7 Sparse Region Handling

Sparse semantic regions require backoff mechanisms. A hybrid resolution rule may be

$$P(c \mid x) = \lambda P_{topo}(c \mid x) + (1 - \lambda) P_{embed}(c \mid x)$$

When $\rho(c)$ is low, systems may reduce λ, expand neighborhood radius k, or retain multi-coordinate uncertainty rather than forcing premature resolution.

7.8 Complexity Considerations

Let

$$|C_x| = k$$

and average neighborhood size m.

Naive resolution cost:

$$O(km)$$

h-hop traversal cost:

$$O(km^h)$$

Practical systems therefore require locality bounds, such as $h \leq 2$, degree bounds, such as $|\mathcal{N}(c)| \leq d_{max}$, candidate pruning, lexical priors, approximate retrieval, and cached resolutions to maintain tractability.

SCIT introduces a shift from stateless semantic reconstruction to stateful semantic infrastructure. Coordinates accumulate disambiguation decisions over time, allowing future inference to operate over previously stabilized semantic identities rather than re-solving equivalent ambiguity. This converts semantic interpretation into a partially persistent computational resource.

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8. Embeddings and Coordinate Realization

SCIT is not a symbolic replacement for neural representations. Coordinates remain compatible with vector-space learning.

Define

$$\phi : V \to \mathbb{R}^n$$

mapping coordinates into embedding space. The desired constraint is

$$d_{topo}(c_i, c_j) \approx d_{embed}(\phi(c_i), \phi(c_j))$$

Topological proximity should therefore correspond approximately to embedding proximity.

Because topological and embedding distances may operate on different scales, distance normalization or learned scale alignment is required. For example

$$\tilde{d}_{topo} = \frac{d_{topo} - \mu}{\sigma}$$

A topology-preservation loss may be defined as

$$\mathcal{L}_{topo} = \sum_{i,j} \left| \tilde{d}_{topo}(c_i, c_j) - \|\phi(c_i) - \phi(c_j)\| \right|$$

Potential implementation approaches include graph-Laplacian initialization, spectral embeddings, topology-aware contrastive learning, adjacency-biased attention, and topology regularization during training. SCIT therefore augments embedding systems rather than replacing them.

8.1 Coordinate-Aware Attention

Coordinate-aware attention may incorporate typed relation bias. For coordinates c_i, c_j,

$$Attn(c_i, c_j) = \text{softmax}_j\left(q_i^\top k_j + b_{\tau(e_{ij})}\right)$$

where $b_{\tau(e_{ij})}$ is an attention bias derived from relation type. This allows topology to influence transformer computation without replacing learned attention.

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9. Retrieval and Addressability

A semantic coordinate system must support sublinear-time retrieval.

First define neighborhood embedding aggregation

$$\phi(\mathcal{N}(c)) = \frac{1}{|\mathcal{N}(c)|} \sum_{c' \in \mathcal{N}(c)} \phi(c')$$

Given query q, retrieval may be defined as

$$\arg\max_{c \in V} \left(sim(q, \phi(c)) + \lambda \cdot sim(q, \phi(\mathcal{N}(c))) \right)$$

This ensures topology participates in retrieval rather than merely embedding similarity. Practical implementations require ANN or equivalent approximate indexing for scalability. The base ANN index key is

$$\phi(c)$$

An augmented index key may be

$$\phi(c) \oplus \phi(\mathcal{N}(c))$$

where $\oplus$ denotes vector concatenation or another governed fusion operator. This connects SCIT naturally to retrieval systems while preserving semantic addressability.

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10. Composition and Structured Semantics

Not all meaning reduces to atomic coordinates. SCIT therefore introduces composition

$$\oplus : V^n \to V \cup \mathcal{G}$$

where $\mathcal{G}$ is the space of structured semantic graphs. Thus

$$c_{comp} \oplus (c_1, c_2, \ldots, c_n)$$

The resulting structure may:

- map to an existing coordinate
- remain a structured semantic expression

- enter SemCrys evaluation for future crystallization

Composition should satisfy associativity where the represented semantic operation permits

$$(c_1 \oplus c_2) \oplus c_3 = c_1 \oplus (c_2 \oplus c_3)$$

The composed structure must satisfy governance validity

$$G(c_{comp}) > 0$$

When $c_{comp} \in \mathcal{G}$, collapse requires graph-to-coordinate projection

$$\psi : \mathcal{G} \to V$$

If there exists coordinate c^* such that

$$d_{topo}(\psi(c_{comp}), c^*) < \epsilon$$

then the composition may collapse to existing coordinate c^*.

Composition must also be cost-controlled

$$C_{comp} \propto n$$

with practical systems bounding

$$n \leq n_{max}$$

For example, legitimate authority under crisis may remain a structured semantic subgraph rather than collapsing into a single coordinate. This preserves generative flexibility while retaining coordinate identity.

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11. Coordinate Compression Economics

SCIT introduces a different notion of compression. The relevant question is not whether fewer characters are transmitted. The relevant question is whether semantic reconstruction overhead is reduced.

Define

$$C_{total} = C_{resolution} + C_{maintenance} - C_{reuse}$$

SCIT is beneficial iff amortized semantic reuse exceeds the cost of maintaining coordinate infrastructure. This converts semantic compression into a measurable engineering tradeoff.

A caching layer may be defined as

$$\mathcal{C}(x) \to c$$

with reuse contribution

$$C_{reuse} \propto |\mathcal{C}|$$

Cache scope may be local to a model, shared across a deployment, or maintained as infrastructure. A cache entry is invalidated when coordinate identity drift exceeds the persistence threshold

$$d_{topo}(c^{(t)}, c^{(t+1)}) > \delta$$

Indirect drift also matters. A cache entry should be invalidated when

$$\exists c' \in \mathcal{N}(c) \text{ such that } d_{topo}(c'^{(t)}, c'^{(t+1)}) > \delta$$

This prevents cached meanings from drifting through neighborhood mutation.

More generally, if semantic reuse frequency $f(c)$ exceeds maintenance cost over time, then the SCIT asymptotic hypothesis is

$$\lim_{t \to \infty} C_{total}^{SCIT} < C_{total}^{lexical}$$

This is not asserted as universally true. It is a testable hypothesis for high-reuse semantic regimes.

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12. Incremental Deployment Path

SCIT does not require immediate replacement of existing tokenizers. A practical deployment path may proceed through:

- Phase 1: Post-hoc coordinate tagging.
- Phase 2: Hybrid lexical-coordinate streams.
- Phase 3: Coordinate-aware attention.
- Phase 4: Fully coordinate-native pretraining.

This progression enables gradual adoption while preserving compatibility with contemporary language models.

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13. Relationship to Existing Systems

Knowledge graphs provide semantic relations but do not define identity through topological stability. WordNet and BabelNet remain lexically anchored rather than coordinate-native. Embedding spaces provide continuity but lack explicit addressability and governance constraints. Wikidata provides identifiers but does not require topological identity semantics.

SCIT may therefore be characterized as:

> addressable, topologically constrained, governance-bound semantic identity.

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14. Semantic Frontier Regions

Define frontier set

$$F \subseteq V \cup V_{proto}$$

representing provisional semantic structures characterized by:

- low stability
- incomplete topology
- unresolved governance consensus
- active semantic evolution

Coordinates within frontier regions may remain probabilistic

$$c \sim P(V_{proto})$$

allowing semantic exploration without premature canonicalization.

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15. Non-Goals

SCIT does not attempt:

- complete ontology closure
- elimination of all ambiguity
- replacement of probabilistic reasoning
- suppression of linguistic plurality
- universal semantic centralization

The framework instead proposes governed coordinate infrastructure for semantic identities that are sufficiently stable, reusable, and topologically integrated to justify representation.

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16. Threat Model and Failure Modes

Assume adversaries may:

- propose coordinates
- modify topology
- bias governance weights

Attack surfaces include:

- semantic drift injection
- coordinate shadowing
- topology poisoning

Robustness objective:

$$\min \mathit{impact}_{adv}(V, E)$$

Additional risks include:

- semantic aliasing
- sparse-region instability
- governance capture
- topology maintenance cost
- coordinate proliferation
- semantic ossification

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17. Minimal Implementation Substrate

A minimal SCIT implementation may consist of:

- Coordinate Registry: coordinate identifiers, namespaces, governance metadata, lineage tracking.
- Topology Store: adjacency lists, typed edges, weighted relations.
- Resolver: candidate generator, topology-conditioned scoring function, governance filter.
- Embedding Layer: topology-regularized embeddings, coordinate-aware attention.

This architecture is sufficient to construct an experimental SCIT prototype.

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18. Empirical Evaluation Program

SCIT should be evaluated empirically rather than accepted conceptually.

Tasks:

- word-sense disambiguation
- multi-hop reasoning
- cross-lingual semantic alignment

Metrics:

- resolution consistency
- attention entropy reduction
- compute per resolved concept
- topology coherence
- embedding-topology correlation

Importantly, negative outcomes remain possible. SCIT may underperform in highly creative language, rapidly evolving domains, or low-redundancy semantic environments. Such results would represent meaningful evidence rather than failure of the scientific process.

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19. Discussion

The dominant assumption of language modeling has been that lexical tokenization is the natural interface between language and cognition. SCIT challenges this assumption.

The framework suggests that the true computational object may not be the token, nor the word, nor even the phrase. Instead, the fundamental cognitive object may be the semantic coordinate.

If so, current tokenization systems may represent an intermediate historical stage rather than an endpoint. Future systems may increasingly organize themselves around governed semantic address spaces whose coordinates are directly consumable by reasoning architectures.

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20. Conclusion

SCIT proposes a shift from lexical tokenization toward semantic coordinate tokenization. Rather than repeatedly reconstructing meaning from ambiguous lexical surfaces, reasoning systems may operate over governed semantic coordinates that function simultaneously as identity, address, and topological location.

The framework extends semantic crystallization into semantic addressability. Its central proposition is simple:

> Meaning becomes more computationally tractable when concepts possess stable semantic coordinates.

The future of machine cognition may therefore depend not only upon larger models or larger context windows, but upon the emergence of governed semantic coordinate systems capable of organizing cognition around meaning itself.

Appendix A – SCIT Stack

- Lexical Space (L)
- ↓ GTK Resolution
- ↓ Semantic Coordinate Space (V)
- ↓ SCIT
- ↓ Embedding Space ($\mathbb{R}^n$)
- ↓ Reasoning System

Appendix B – Coordinate Properties

Every coordinate should possess:

- uniqueness
- topological location
- adjacency structure
- contrastive boundaries
- governance metadata
- revision lineage
- jurisdictional scope
- retrieval addressability

Appendix C – Future Research Directions

- coordinate allocation algorithms
- topology-aware embedding initialization
- coordinate-native transformers
- semantic routing systems
- coordinate compression economics
- multilingual coordinate projections
- semantic governance protocols
- topology-aware reasoning benchmarks

Related Works

- semantic coordinate tokenization
- topological identity
- governed semantic knowledge
- semantic disambiguation
- coordinate-native representation
- semantic addressability
- semantic crystallization
- topology-aware attention
- semantic governance
- cross-lingual coordinate projection
- semantic compression economics
- coordinate-indexed tokenization

Corpus-Wide Reference Positioning

This paper argues that large language models currently operate over tokenization systems optimized around lexical recurrence, statistical compression, and surface-level representational efficiency. Its nearest external comparison fields are semantic, token, crypto. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: Formal vocabularies and graph links make commitments explicit but cannot by themselves establish correct identity, shared interpretation, or lossless translation between contexts. Statistical compression can improve efficiency while destroying distinctions needed for interpretation; semantic labels can conversely overstate stability that usage does not support. A valid hash, signature, or inclusion proof establishes a bounded cryptographic relation, not meaning, authority, safety, or jurisdiction by itself. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] Richard Cyganiak, David Wood, and Markus Lanthaler, eds. *RDF 1.1 Concepts and Abstract Syntax*. W3C Recommendation, 25 February 2014. https://www.w3.org/TR/rdf11-concepts/ **Relationship:** Adjacent graph data model and identity baseline.

[**E2**] Alistair Miles and Sean Bechhofer, eds. *SKOS Simple Knowledge Organization System Reference*. W3C Recommendation, 18 August 2009. https://www.w3.org/TR/skos-reference **Relationship:** Adjacent standard for concepts, labels, schemes, notes, and semantic mappings.

[**E3**] W3C OWL Working Group. *OWL 2 Web Ontology Language Document Overview*, Second Edition. W3C Recommendation, 11 December 2012. https://www.w3.org/TR/owl2-overview/ **Relationship:** Adjacent formal ontology standard used to compare declared semantics and inference commitments.

[**E4**] Mark D. Wilkinson et al. "The FAIR Guiding Principles for Scientific Data Management and Stewardship." *Scientific Data* 3, 2016, 160018. DOI: 10.1038/sdata.2016.18. https://doi.org/10.1038/sdata.2016.18 **Relationship:** Predecessor stewardship principles for findability, accessibility, interoperability, and reuse.

[**E5**] Peter Gardenfors. *Conceptual Spaces: The Geometry of Thought*. MIT Press, 2000. mitpress.mit.edu source **Relationship:** Predecessor geometric treatment of concepts; cited as comparison, not as validation of the paper's manifold terminology.

[**E6**] Rico Sennrich, Barry Haddow, and Alexandra Birch. "Neural Machine Translation of Rare Words with Subword Units." *ACL 2016*, pp. 1715-1725. DOI: 10.18653/v1/P16-1162. https://doi.org/10.18653/v1/P16-1162 **Relationship:** Predecessor subword tokenization method used as a contrast to semantically governed token boundaries.

[**E7**] Taku Kudo and John Richardson. "SentencePiece: A Simple and Language Independent Subword Tokenizer and Detokenizer for Neural Text Processing." *EMNLP 2018: System Demonstrations*. DOI: 10.18653/v1/D18-2012.

https://doi.org/10.18653/v1/D18-2012 **Relationship:** Predecessor tokenizer implementation used to separate statistical segmentation from semantic admissibility.

Ableman corpus

[**A1**] Adam Ableman. *Meaning as Binding.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-5F976917F0B283EA`;
source SHA-256
`A470D1A0B9ED7C3CEAB6BB88BE4B642A2BA36427447F787764329FD574DA0C97`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Governed Meaning Base (GMB).* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-D95CCC4FB7B58BFB`;
source SHA-256
`A3B62E50F8BEE125FDAE0D38C2839AFD1CCC138296268BF93217481E38AA48EA`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Semantic Manifold Topology.* Ableman Corpus current edition; work identity `PAPER.PUB.20257244`;
source SHA-256
`B8F28CC652A73A3EE4CF6FD0E23C1ABC922A6660319B2FB4CC1CCAAA606EC64A`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Global Term Key: Identity Primitives in Governed Knowledge Systems.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-358AC09F5A13587E`;
source SHA-256
`AFB29CFCB4B58A78B79709AD453CEE9088CCB2AE40F30A70260F1B54F5160579`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Semantic Legitimacy Before Context.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-091E34BA93992CE4`;
source SHA-256
`DEB7626CE1AB54F4B541347DCC3C4E14DC4F682230ACCCE1A93D1D536A50C44C`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *Governed Semantic Tokens: Tokenization as an Admissibility Boundary.* Ableman Corpus current edition; work identity `PAPER.PUB.18469507`;

source SHA-256
`5C44175B87C514892A0DC9743E09D03893E970AF5419F2E6D9171651BE2A9A54`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *Token Parsimony.* Ableman Corpus current edition; work identity `PAPER.PUB.20382825`;
source SHA-256
`EC3AA22D32E2BBA21CFCB6D1FA2AE6EA8AC11B7781D19C51530A2F6D924B2E91`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Editorial source notes retained from predecessor

Ableman, Adam. *SemCrys: Toward a Tokenizer-Native Semantic Substrate for Machine Cognition.*

Ableman, Adam. *Global Term Key (GTK).*

Bengio, Yoshua et al. "Representation Learning."

LeCun, Yann. "Path Towards Autonomous Machine Intelligence."

Leibniz, Gottfried Wilhelm. *Characteristica Universalis.*

Mikolov, Tomas et al. "Distributed Representations of Words and Phrases and their Compositionality."

Tenenbaum, Joshua et al. "How to Grow a Mind."

Wilkins, John. *An Essay Towards a Real Character and a Philosophical Language.*

WordNet Project.

Wikidata Knowledge Graph.

BabelNet.

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
License: CC-BY-ND-4.0

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